# Peer review and response

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

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WORKBOOK · 1 OF 15

## Recall and choose your starting point

> Record your starting explanation for Peer review and response before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Writing results and limitations

Without opening the old lesson, explain one key idea from Writing results and limitations. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Figures that preserve evidence

Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Inspecting failures and anomalies

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 15

## Practise with fading help

> Complete the supported Peer review and response trace and record the first step you can explain without help.

Use W2: Trace the worked case for Peer review and response. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

## Individual reasoning check

> Check the proposed Peer review and response explanation against the supplied facts before discussing it.

Use the worked case for Peer review and response. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 15

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Apply a new case independently

> Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 15

## W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Peer review and response facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**My reasoning and evidence**

WORKBOOK · 6 OF 15

## W2 Trace the worked case

> Trace the worked case. Apply the supplied Peer review and response facts and keep your reasoning.

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

WORKBOOK · 7 OF 15

## W3 Audit the limit

> Audit the limit. Apply the supplied Peer review and response facts and keep your reasoning.

A report omits this qualification: Agreement with a reviewer is not required, but a disagreement needs explicit reasoning and **evidence**.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

WORKBOOK · 8 OF 15

## W4 Apply a new case independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Peer review and response facts and keep your reasoning.

Reviewer notes missing failure counts; author deletes the limitation paragraph. What is an adequate response?

**My reasoning and evidence**

WORKBOOK · 9 OF 15

## W5 Design a revealing follow-up

Design a revealing follow-up. Apply the supplied Peer review and response facts and keep

> your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

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WORKBOOK · 10 OF 15

## Project checkpoint and reflection

> Connect Peer review and response to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled peer review and response **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of peer review and response.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

## Pictorial case practice: Peer review and response

> Respond to a reviewer with evidence. A reviewer asks why four photos of one plant were treated as four independent plants. Explain the labelled step before checking the changed condition.

Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

The original table contains four views of P1 and four views of P2: eight photos from two plants.

AI ACADEMY / AI-7/32

# Peer review and response

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Identify the concern

The reviewer questions the unit of analysis.



## 2. Check the records

Plant identifiers confirm two plants, not eight independent plants.



## 3. Revise transparently

Correct the description and revise any analysis that assumed independence.

**CHANGE ONE THING**

The team only changes "photos" to "plants" without revisiting analysis.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Count photos and underlying plants in the reviewer's concern. What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 14 OF 15

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Correct the description and revise any analysis that assumed independence.

1. Revise transparently

2. Check the records

3. Identify the concern

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Revise transparently Correct the description and revise any analysis that assumed independence.

Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: The team only changes "photos" to "plants" without revisiting analysis. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The substantive independence issue remains unresolved.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The substantive independence

issue remains unresolved. Renaming records does not change their shared source objects.

### Repair a missing building block

Review Writing results and limitations (ai-7/31). Goal: Explain the **mechanism** and terminology of writing results and limitations. Short practice: Compute the supported-answer rates for 9/12 and 10/12.

What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?. Return to this week and explain what you changed.

Review Figures that preserve **evidence** (ai-7/29). Goal: Explain the **mechanism** and terminology of figures that preserve **evidence**. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

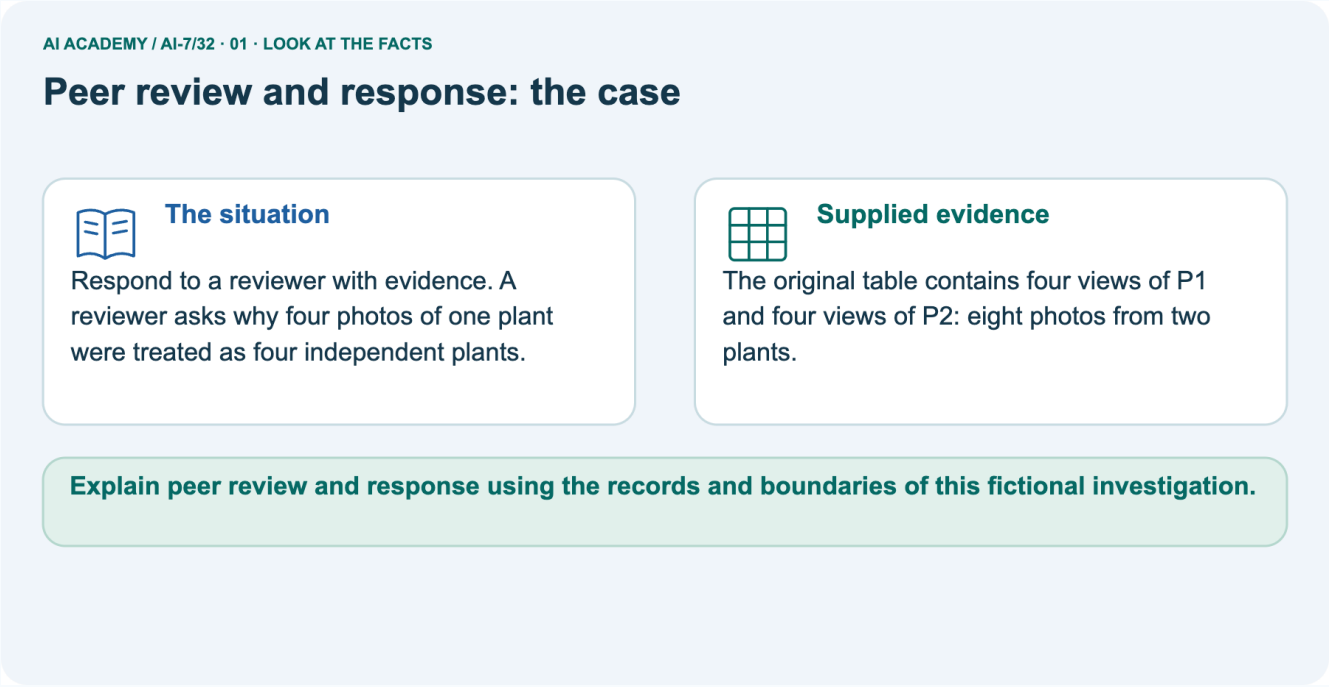
Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

## Visual concept journey

> Explain each Peer review and response picture from its labels, then record a relevant limit.

### 01. Peer review and response: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

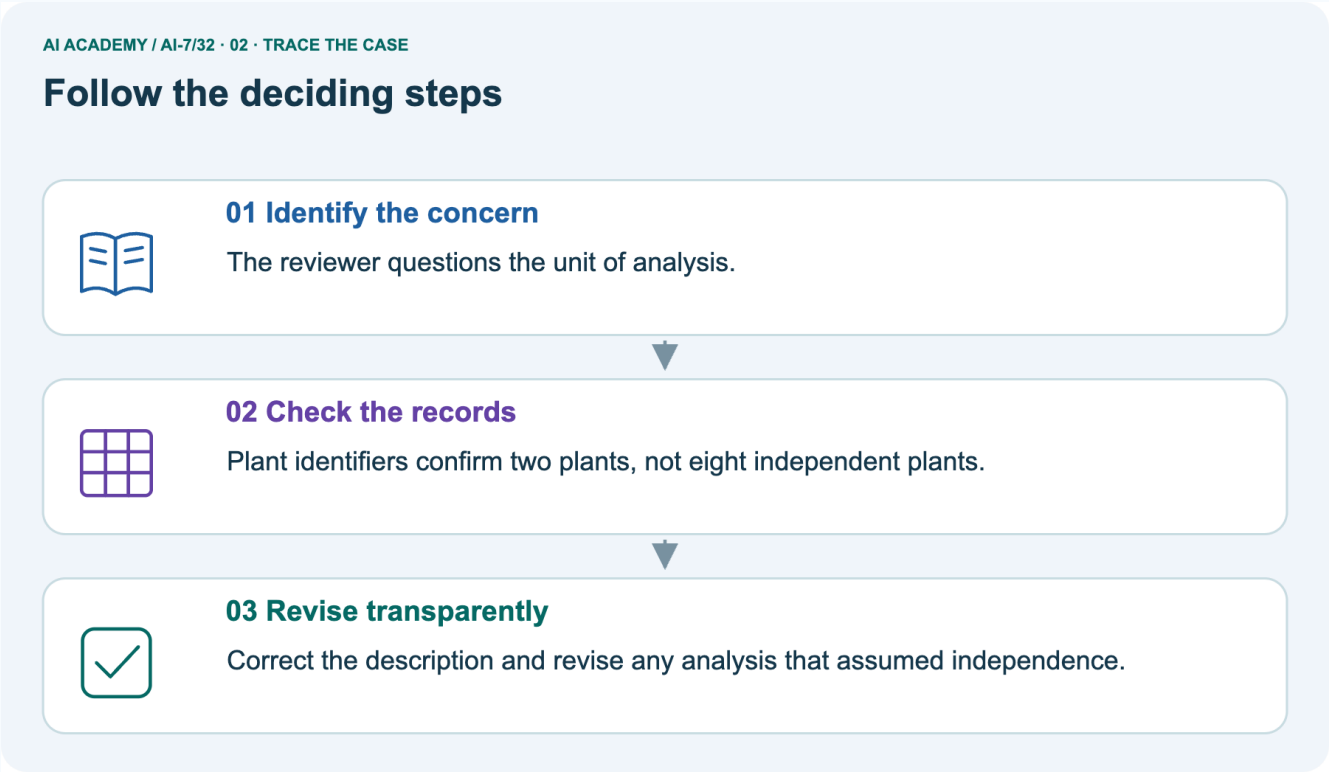
Respond to a reviewer with **evidence**. A reviewer asks why four photos of one plant were treated as four independent plants.

Supplied facts: The original table contains four views of P1 and four views of P2: eight photos from two plants.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



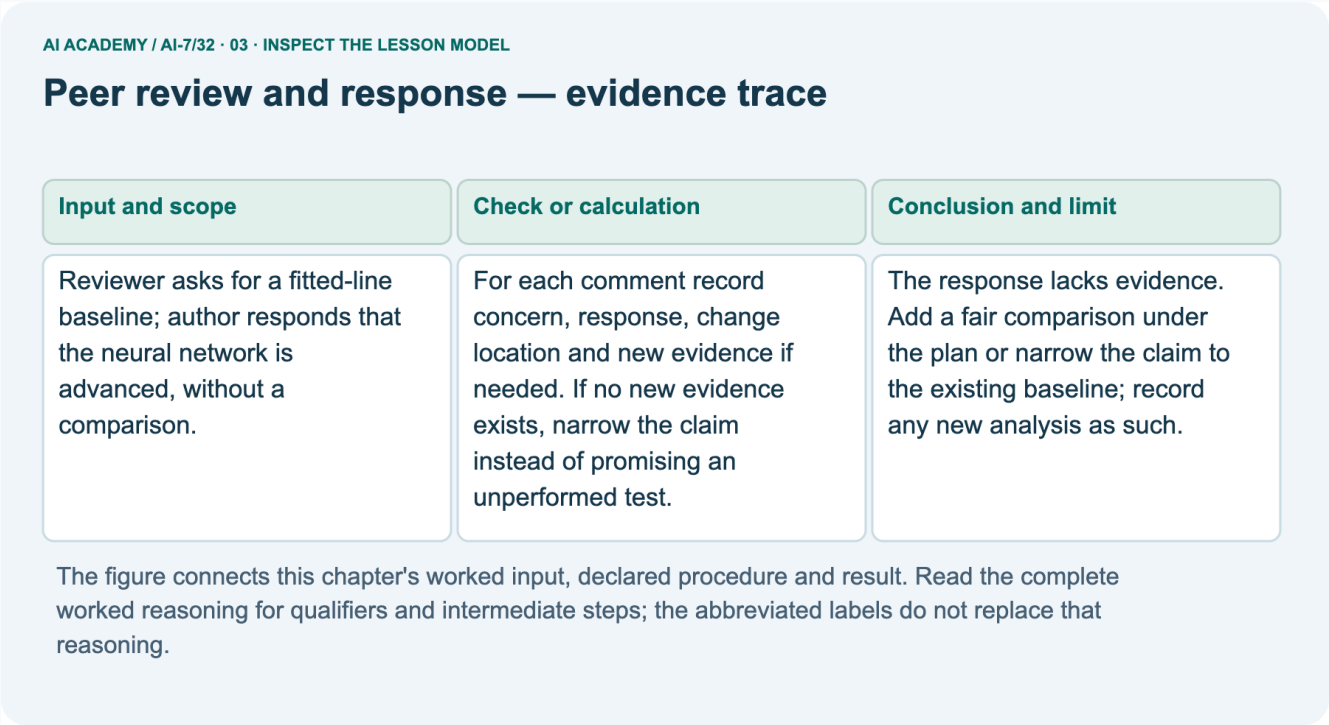

Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the concern: The reviewer questions the unit of analysis. → Check the records: Plant identifiers confirm two plants, not eight independent plants. → Revise transparently: Correct the description and revise any analysis that assumed independence.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Peer review and response — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

Peer review and response — **evidence** trace.

Reviewer asks for a fitted-line baseline; author responds that the neural network is advanced, without a comparison. | For each comment record concern, response, change location and new **evidence** if needed.

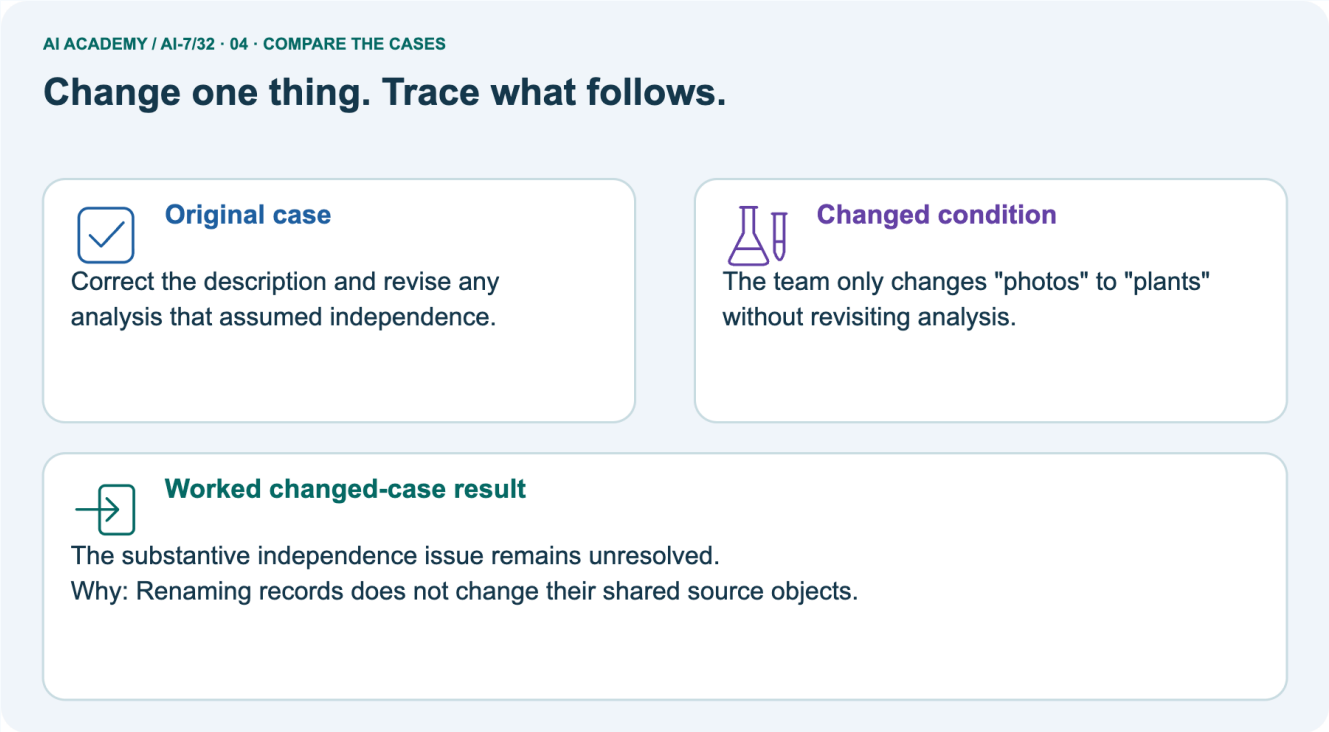
If no new **evidence** exists, narrow the claim instead of promising an unperformed test. | The response lacks **evidence**.

Add a fair comparison under the plan or narrow the claim to the existing baseline; record any new analysis as such.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

**04. Change one thing. Trace what follows.**




Follow the labelled picture with the matching explanation. Open the vector diagram

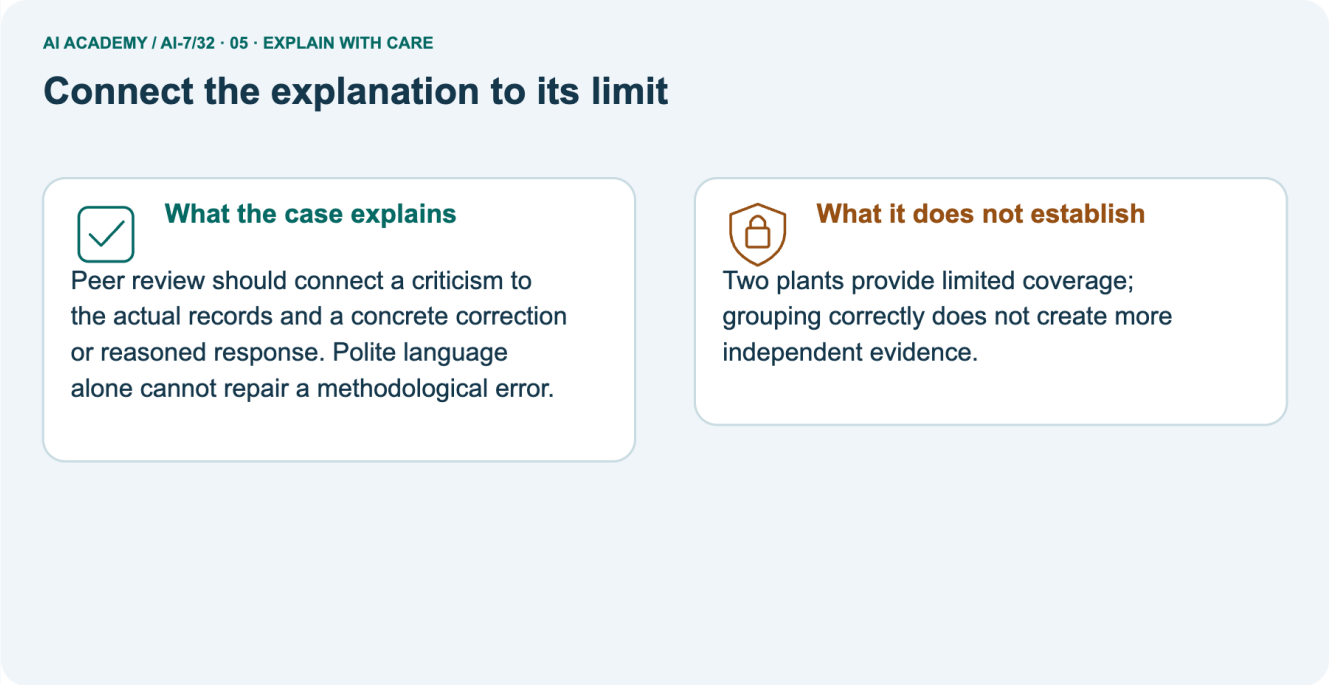
Original case: Correct the description and revise any analysis that assumed independence.
Changed condition: The team only changes "photos" to "plants" without revisiting analysis.

Expected result: The substantive independence issue remains unresolved. Reason: Renaming records does not change their shared source objects.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

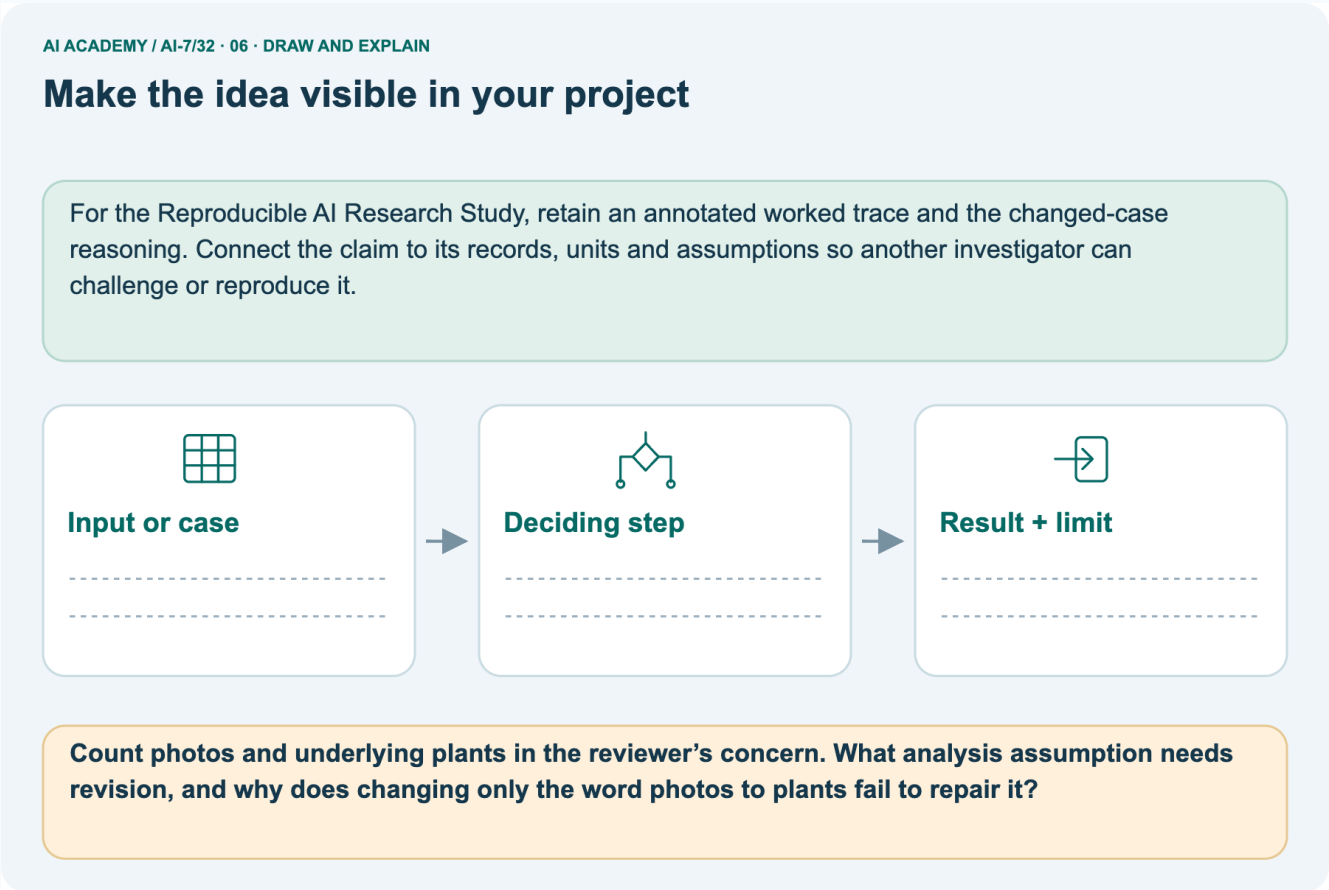
Peer review should connect a criticism to the actual records and a concrete correction or reasoned response. Polite language alone cannot repair a methodological error.

Limit: Two plants provide limited coverage; grouping correctly does not create more independent **evidence**.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Count photos and underlying plants in the reviewer’s concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?

> Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.